

Breakthrough in cooling?

There's a new method of making liquid flow vertically along a silicon surface without a pump or any similar mechanism. you can get a look at the latest in CPU cooling technology at <http://bit.ly/cf8SKP>

New ASUS P55 mobo

ASUS has announced a new, ultra-enthusiast option called the Maximus III Extreme that is reportedly costlier than some X58 options

How stuff works



Evergreen?

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AMD has good reason to cheer over the past seven months or so. Quite a turnaround from the past couple of years that saw both Intel and NVIDIA sticking it to them repeatedly. AMD's Radeon HD 2xxx series was a downright failure, the HD 3xxx that came soon after bombed badly as well. In a chat we had with Richard Huddy, Senior Manager Developer Relations AMD, at the launch of the Radeon HD 5xxx series last year he said "the 3xxx was actually the starting point for the really strong HD 4xxx series, and it is the HD 4870 that led us here". Evidently they were on the right track with

the direction the architecture and their design ethos was taking.

A brief look at the family

Evergreen is the codename for AMD's DX11 family. AMD has a full range of DX11 cards, and the introduction of Fermi to the market notwithstanding, AMD is in a strong position because they have DX11 solutions to suit every budget and have had some seven months to iron out inevitable teething issues. The first cards to launch were the Radeon HD 5870 and HD 5850, codenamed Cypress. Then came Hemlock, the HD 5970 — a dual GPU option, consisting of two HD 5870 GPUs, but at lower clock speeds. These are the top-end parts and soon after there came a slew of scaled-down parts.

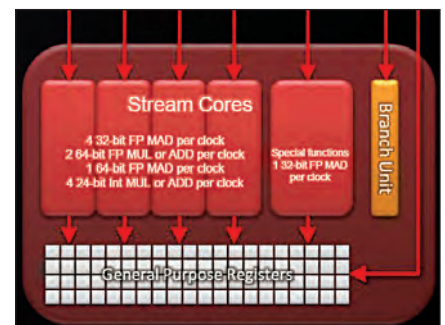
Meet the family

Codename	Card Name	No of ALUs	Memory / Bus	Price Point
Hemlock	HD 5970	1600 x 2	2 GB, GDDR5 / 256-bit x2	Rs. 40,000
Cypress	HD 5870 / HD 5850 / HD 5830	1600 / 1440 / 1120	1 GB GDDR5 / 256-bit	Rs. 26,000 / 17,000 / 13,000
Juniper	HD 5770 / HD 5750	800 / 720	GDDR5 / 128-bit	Rs. 10,000 / 8,500
Redwood	HD 5670	400	GDDR5 / 128-bit	Rs. 6,500
Redwood	HD 5570	400	GDDR3 / 128-bit	Rs. 4,700
Cedar	HD 5450	80	GDDR2/GDDR3	Rs. 3,500

Last month we looked at NVIDIA's shiny new DX11 offering. AMD has been in this segment for quite a while now, so it's only fair we take a look at the GPU family that almost single-handedly restored their dwindling fortunes

A look at the GPU

The most touted term in AMD's jargon box has to be *TeraScale 2*. This simply means the RV870 (the codename for the HD 5870 GPU) is capable of providing over two teraflops of processing power (2.7 teraflops to be exact), which is a 2x hike over the RV770 (Radeon HD 4870). There's a lot more hierarchy to the GPU



One thread processor consists of 4 ALUs and one SFU (Special Function Unit)

block design compared to predecessors. This is obviously required thanks to the doubling of the ALU count. A modular architecture is the only way to go for scalability. The 1600 SPs on the HD 5870 are thus divided into 20 SIMD engines. Each SIMD engine has 16 thread processors. Each thread processor has five stream cores. This looks pretty similar to the RV770 (HD 4870). But there are a number of small under-the-hood changes.

One thing remains unchanged. ATi's thread processor still consists of four ALUs and one special function unit, and therefore doesn't consist of 5 SPs in the truest sense of the term. The ratio